COMPUTATIONAL COMMUNICATION RESEARCH 8.1 (2026) –
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Appendix A Outlet list

Both major legacy and alternative outlets were included. The latter outlets are generally characterized by a self-understanding as a "corrective, opposing the overall tendency of public discourse emanating from what is perceived as the dominant mainstream media in a given system" (Holt et al., 2019, p. 3) (see also Schwaiger (2022)). They more or less explicitly claim to publish and promote content that is unduly underrepresented in the mainstream public sphere. We included major alternative news outlets in terms of reach, but also of differing political leaning, hence offering a broad understanding of alternative media as a corrective to a biased mainstream media system.

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Outlet	URL	Twitter handle
BILD	bild.de	@BILD
Sueddeutsche Zeitung	sueddeutsche.de	@SZ
Frankfurter Allgemeine Zeitung	faz.net	@faznet
Handelsblatt	handelsblatt.com	@handelsblatt
Die Welt	welt.de	@welt
Focus	focus.de	@focusonline
Zeit (Online)	zeit.de	@zeitonline
Frankfurter Rundschau	fr.de	@fr
Stern	stern.de	@sternde
TAZ	taz.de	@tazgezwitscher
Spiegel	spiegel.de	@derspiegel
Epoch Times	epochtimes.de	@EpochTimesDE
PI-NEWS	pi-news.net	@P_I
Journalistenwatch	journalistenwatch.com	@jouwatch
Die Achse des Guten	achgut.com	@Achgut_com
Tichys Einblick	tichyseinblick.de	@TichysEinblick
Junge Freiheit	jungefreiheit.de	@Junge_Freiheit
Deutsche Wirtschafts Nachrichten	deutsche-wirtschafts-nachrichten.de	@DWN_de
NachDenkSeiten	nachdenkseiten.de	@NachDenkSeiten
Cicero	cicero.de	@cicero_online
Neues Deutschland	nd-aktuell.de	@ndaktuell
Der Freitag	freitag.de	@derfreitag
junge Welt	jungewelt.de	@jungewelt
netzpolitik	netzpolitik.org	@netzpolitik_org
Correctiv	correctiv.org	@correctiv_org
Uebermedien	uebermedien.de	@uebermedien

Appendix B List of MPs

The basis for the collection was a list of 629 MPs in parliament for the federal legislature starting 2021 with registered Twitter accounts provided by the *Epinetz* data set (König et al., 2022). The follower network was then collected for all MPs that were actively participating in public debate in the German Twittersphere. To this end, we included all MPs that showed up in the retweet networks of at least 5 of around 2000 Twitter trending topics, collected daily from 2021 to 2023. This yielded 433 MPs. The follower network was collected in July 2023. For a list of these MPs, see separate `.csv` file.

Appendix C Sample

Since the sample consists of three different yet interrelated data collections, we briefly sum up what data the pipeline relies on:

1. The Twitter follower network of (in this case 433 German) MPs, collected with the Twitter API v1.[1] We only included users following at least three MPs in our analysis, which resulted in 1.2 million followers being included in our pipeline.
2. All tweets from March 2023 including a link to at least one of the 26 outlets (see Appendix). Collection was carried out with the Twitter Research API at the beginning of April 2023 and resulted in roughly 500,000 unique tweets. The data set also includes shortened links (such as `bit.ly/AAAAA`). Sharing events can only be included in the analysis if the users sharing the news also appear in the embedding, i.e. follow at least three MPs. Roughly 60 percent of all shares in the sample came from embedded users.
3. The full texts of all articles that were not paywalled. 75 percent of all shares could be assigned a text (60,000 unique articles). Text collection was carried out with the `newspaper3k` Python library (Ou-Yang, 2020).

[1] For follower collection, the outdated v1 API access provided better performance than the Research API.

Appendix D Correspondence Analysis

The follower network can be described as a adjacency matrix A with N_{MP} columns and N_{users} rows. An entry a_{ij} is 1 if a user follows the respective MP, and 0 otherwise. The correspondence matrix P is calculated by $P = \frac{A}{\sum_{i,j} a_{ij}}$. The matrix of standardized residuals S is calculated via

$$S = D_r^{1/2}(P - rc^T)D_c^{1/2},$$

with r and c being row and column masses ($r_i = c_j = \sum_j p_{ij}$, $D_r = \text{diag}(r)$, $D_c = \text{diag}(c)$) (see Barberá et al. (2015), Sup. Mat.). Then, Singular Value Decomposition is carried out such that $S = UD_aV^T$ where $U^TU = V^TV = \mathbb{I}$. All rows and columns are projected onto that plane by then computing the standard coordinates $D_c^{1/2}V$ for MPs, $D_r^{1/2}U$ for other users (Barberá et al., 2015).

Using CA (Barberá et al., 2015; Bonica, 2014; Ramaciotti Morales et al., 2022) is computationally more efficient than using latent space models (Barberá, 2015; Hoff et al., 2002).[2] In the present case, with more than one million users being embedded, we hence choose CA. The closer a user and an MP are placed together in the space produced by CA, the higher the probability that the user follows the MP tends to be. We interpret the space as a political space and users placed close together as users sharing a political position.

As in Ramaciotti Morales et al. (2022), we only included users that followed at least three MPs. But we take the full user-MP follower matrix, and not a reduced full-rank matrix. The reason for this is that a latent space inferred by only including unique connection patterns can turn out to be different from the original one, which becomes intuitively clear in a force-based formulation of latent space models: More users with the same following pattern increase the attracting force on the MPs they follow, and the repulsion on the others (see Gaisbauer et al. (2023)).

Percent of variance explained is displayed in Fig. 1.

We note here that we leave out the second Principal Component (PC) of the CA and operate with PCs 1 and 3. Along the second PC, MPs of the different parties are spread out far more widely than along the other axes (for a graphical display, see Appendix). Along dimensions 1 and 3, the MP positions of the different parties never have a variance larger than 0.2, while *all* except one party has a variance larger than 0.2 along dimension 2. We find

[2]Even though both approaches are often employed in computational social research and CA has been validated *with* latent space models as a baseline.

that the dimension correlates with the MPs' follower counts (Spearman Rank 0.79, Pearson 0.58) and thus rather reflects the MPs' popularity.[3] We hence leave out the second dimension because it is not politically meaningful.

Dimensions 1 and 2 of the CA are displayed in Fig. 2.

For each political party in parliament, we take its position as the average of the positions of its MPs. We then rotate the Principal Components of the space and calculate Pearson correlation with all possible orthogonal combinations of the left-right with the other 32 suitable CHES issues of German political parties.[4] In fact, we find that four CHES issues have a nearly equally high maximum correlation (all > 0.91). These are: elite-skepticism (called 'people vs. elite' in CHES), general position towards EU integration, position on EU's internal market and trade protectionism. We provide the detailed CHES description of these issues in the Appendix. The four CHES issues' corresponding optimal rotations are all within a window of only 22 degrees. Hence, we build an issue-combined second axis by averaging over their rotations of maximum correlation. Even with this averaged rotation, Pearson correlation with each of the issue dimensions is higher than 0.9 each.[5]

We center the space around the mean user position along each dimensions (indicated by 0).

The code for the CA and the rotation can be found under `github.com/fgais/political-s`

[3]While it has been claimed that the normalization step in the CA amounts to something similar to the popularity parameters in latent space models (Bonica, 2014), CA apparently does at least not adequately factor out popularity of the MPs in this case.

[4]The CHES includes more issues, but we exclude those that express salience (that is, how important an issue is for a party).

[5]We tested inclusion of arbitrarily many CHES issues with a cutoff of a Pearson correlation of 0.8. While such a fixed cutoff is still arbitrary, we aim to err on the safe side here: It allows us to safely assume that all four CHES issues are sufficiently well-represented by the second dimension, and its interpretation is not diluted by including as many CHES issues as possible.

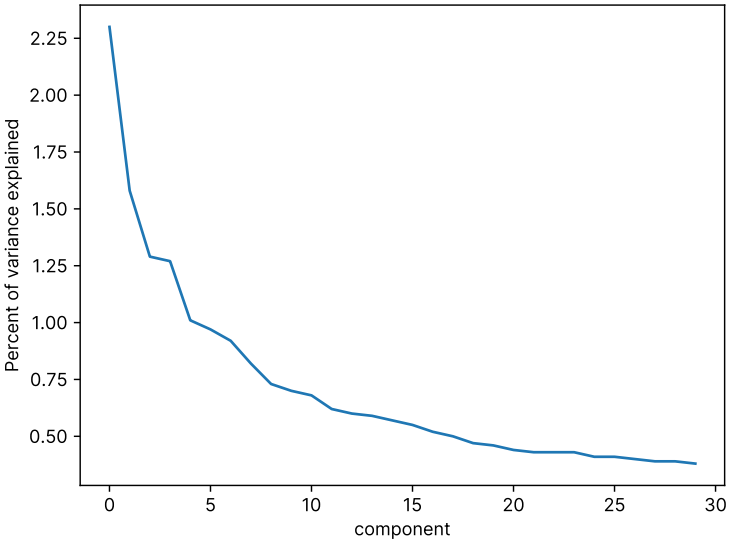

Figure 1: CA percent of variance per dimension.

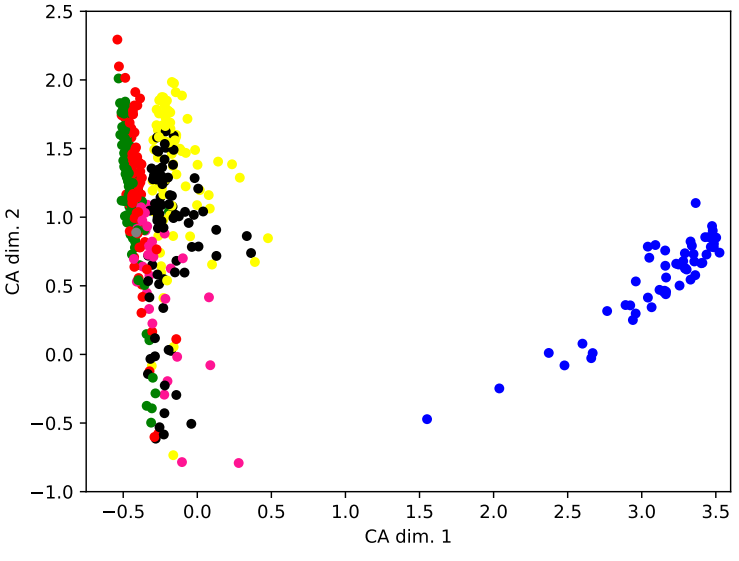

Figure 2: CA first and second dimension: MP positions for each party are disproportionately spread out along the second dimension, and the most important MPs (e.g., the party chairs or the ones with the most followers) tend to be placed close together, no matter the party.

Appendix E CHES codebook entries

- People vs. elite: "[P]osition on people vs elected representatives. Some political parties take the position that 'the people' should have the final say on the most important issues, for example, by voting directly in referendums. At the opposite pole are political parties that believe that elected representatives should make the most important political decisions."
- EU position: "[O]verall orientation of the party leadership towards European integration in 2019"
- EU internal market: "[P]osition of the party leadership in 2019 on the internal market (i.e. free movement of goods, services, capital and labor)."
- Protectionism: "[P]osition towards trade liberalization/protectionism."

All from Jolly et al. (2020).

Regarding the combination of issues in the second dimension, we could also just use the CHES issue with the *highest* correlation to explain the dimension. But the other CHES issues will exhibit high correlation with the dimension even if we ignore them; including them hence can give the axis more explanatory power. This is especially useful considering that CHES asks for both high-level ideological dimensions (e.g., left-right) and more specific issues (anti-islam rhetoric). A combination of the latter gives the interpretation of the dimension more breadth, which is desirable if we operate in a two-dimensional space.

Appendix F Unretrieved articles

See Fig. 3.

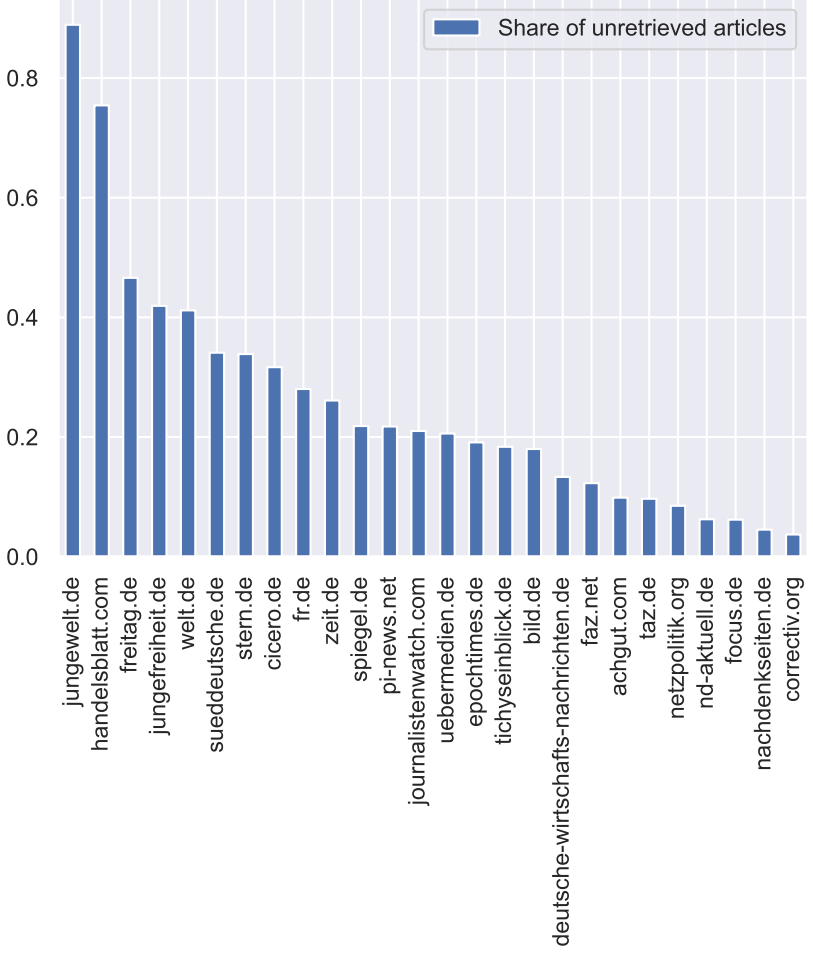

Figure 3: Share of unretrieved articles per outlet.

Appendix G Topic model selection

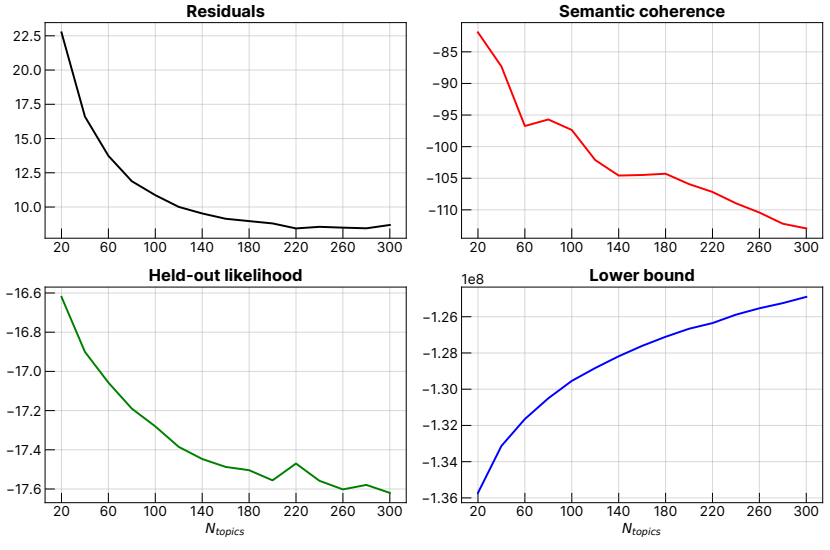

Figure 4: Topic model evaluation scores.

The preprocessing of the raw articles consists of stop-word and URL removal. Then, the articles are tokenized and lemmatized using the Python library `spaCy`, treating named entities and compound nouns as single tokens. The sentence "John Doe ran out of dishwashing soap" thus would turn into "[john_doe,run,dishwashing_soap]". After preprocessing, we infer a structural topic model using the R library `stm`. The important hyperparameter here is the number of topics N_{topics}, which we vary between 20 and 300 in steps of 20, as can be seen in Fig.4. We then evaluate the different models according to standard model evaluation scores: residuals, semantic coherence, held-out likelihood and lower bound. None of these scores should be considered alone to select N_{topics}. Instead, they point the researcher to sensible parameter regions to examine manually. Considering the inflection points in semantic coherence and held-out-likelihood in Fig.4, we manually examined the models for 80, 100 and 220 topics. The latter yielded the best tradeoff between semantically coherent and fine-grained topics, which led us to choose it for the further analysis.

We chose an STM because embedding-based approaches such as `BERTopic` (Grootendorst, 2022) extract topics by clustering documents in a semantic space given by pre-trained sentence embedding models. Such approaches

are well suited for short, idiosyncratic documents such as social media posts, where one usually assumes a single topic per document (Egger & Yu, 2022; Pournaki et al., 2025). In particular, sentence embedding models generate the embeddings via mean-pooling, which can blur the representation of longer documents containing a variety of subtopics. Additionally, the density-based clustering algorithm may classify such documents as "outliers" if their resulting vector lies in between two dense topical clusters. This limits the effectiveness of embedding-based approaches for the analysis of longer texts. In contrast, LDA-based methods such as `stm` explicitly model documents as mixtures of multiple latent topics while relying on stable word co-occurrence patterns that only emerge reliably in sufficiently long text. This makes them naturally suitable for extracting topics from newspaper articles.

This issue could be circumvented if it was possible split the longer article texts into paragraphs each devoted to one topic. But this is not possible in a meaningful way. Some news outlets include paragraphs in their articles for which we might assume that this is the case, but others do not, and hence, we would have to split articles relatively arbitrarily into paragraphs if we were to apply such an approach consistently across all articles.

Appendix H Codebook for metatopics

The metatopics themselves are based on (Quandt, 2008) and (Newman et al., 2023). Coding was done independently for each topic by two coders, who, in case they assigned a topic different metatopics, settled on a common metatopic upon discussion. Initial Cohen's Kappa was .69. The topics along with lists of most representative words, as well as a list of their assigned metatopics, can be found in the separate text files provided.

- Environment/Climate Change:
  - Score keywords refer to (extreme) weather events, nature, natural disasters, or climate change
  - 3 most representative articles mention said terms and information related to them, in particular with a focus on the natural implications (as opposed to social implications)
- Culture/Arts/Media:
  - Score keywords refer to cultural events or artists, works of art and/or cultural entertainment such as films, series or books
  - 3 most representative articles mention said terms and information with a focus on their cultural significance (not on celebrity news/gossip)
- Sports:
  - Score keywords refer to sports-related actors/events
  - 3 most representative articles mention sports-related information with a focus on their significance for their sports disciplines (not on celebrity news/gossip)
- Lifestyle/Human Interest/Celebrities:
  - Score keywords refer to stars, cultural and public figures, or lifestyle with entertainment connotation
  - 3 most representative articles provide information that is primarily entertaining or privately useful, e.g. the private life of stars, cultural and public figures, or DIY content
- Crimes/Personal Security:

  - Score keywords refer to criminal events, courtroom converage, or personal security-related issues/events (e.g. accidents)
  - 3 most representative articles mention criminal events and investigations, courtroom coverage, or personal security-related issues

- Economy/Finance/Business:
  - Score keywords refer to businesses, economic actors (that are not directly tied to political institutions/decisions), financial markets
  - 3 most representative articles mention economic or financial actors or developments (that are not directly focussing on political institutions/decisions)

- Local:
  - Score keywords refer to actors, events or issues of local relevance
  - 3 most representative articles mention issues of only local relevance, such as news on city planning or regional events

- Science/Technology/Health:
  - Score keywords refer to technology, science, or health research/news
  - 3 most representative articles focus on scientific or technological developments, science journalism, or health-related news

- National Politics:
  - Score keywords refer to political or state institutions, politicians, or political events such as elections
  - 3 most representative articles focus on national political issues which are directly tied to political/state institutions, politicians, or political events (such as elections)

- International/Foreign Politics:
  - Score keywords refer to international political issues, institutions, but also national political actors or events of other countries than Germany
  - 3 most representative articles focus on international political issues, especially conflicts (such as the war in Ukraine), institutions (such as the UN), but also national politics of other countries than Germany

- Social Affairs:
  - Score keywords refer to societal issues, e.g. topics of education, migration, or employment and poverty (can be political, but not directly tied to political/state institutions)
  - 3 most representative articles focus on societal issues, e.g. topics of education, migration, or employment and poverty (can be political, but not directly tied to political/state institutions)
- History:
  - Score keywords refer to historical events
  - 3 most representative articles focus on a portrayal of historical events in a both informative and entertaining fashion

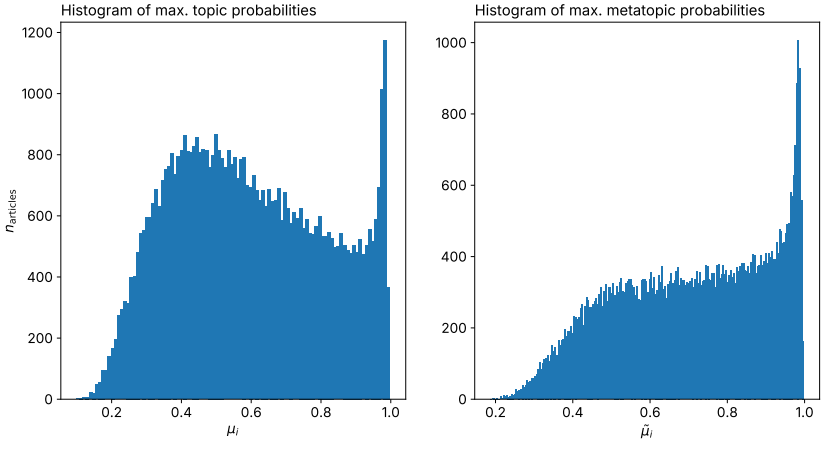

Figure 5: Histograms of max. topic probabilities μ_i (left) and max. metatopic probabilities $\tilde{\mu}_i$ (right).

Appendix I Topic model probability threshold

Let $\mathcal{T} \in \{1, ..., n_{\text{topics}}\}$ be the topic index set. Let $\mathbf{M} = (M_{ij})$ be the document-topic matrix such that M_{ij} is equal to the probability of topic j in document i. After grouping topics into metatopics as described in Sec. 3.3, we define the document-metatopic matrix $\tilde{\mathbf{M}} = (\tilde{M}_{ij})$ such that $\tilde{M}_{ij}$ is equal to the probability of metatopic j in document i. This probability is given by

$$\tilde{M}_{ij} = \sum_{j' \in \mathcal{T}_j} M_{ij'}$$

where $\mathcal{T}_j \subset \mathcal{T}$ is the set of topics that belong to the metatopic j.
In order to examine the distribution of (meta-)topics across the political space in Sec. 3.3, we define a main topic to each document. For each document i, we compute the maximum of its topic probability distribution M_{ij} denoted as μ_i:

$$\mu_i = \max_i(M_{ij}) \tag{1}$$

Equivalently, we define $\tilde{\mu}_i$ as the maximum of the metatopic probability distribution of topic i. Fig. 5 shows the histogram of the values of μ_i and $\tilde{\mu}_i$. For topics (left plot), we observe that the histogram peaks at a value around 0.5. For metatopics (right plot), we observe that the histogram reaches a plateau around a value of 0.5. These observations, combined with a close reading of selected articles with different values of μ_i, leads us to formulate the following rule for assigning the main topic to a document: If $\mu_i > 0.5$,

we assign the corresponding topic as the main topic to document i. Else, we disregard the document for this part of the analysis. Equivalently, we apply the rule $\tilde{\mu}_i > 0.5$ to assign the main metatopic. Following this scheme, we can assign a main topic to 62% of the documents and a main metatopic to 81% of the documents in the corpus. The threshold therefore provides a reasonable tradeoff between accuracy and sample size.

Appendix J Crime/Personal Security news from focus.de

See Fig. 6.

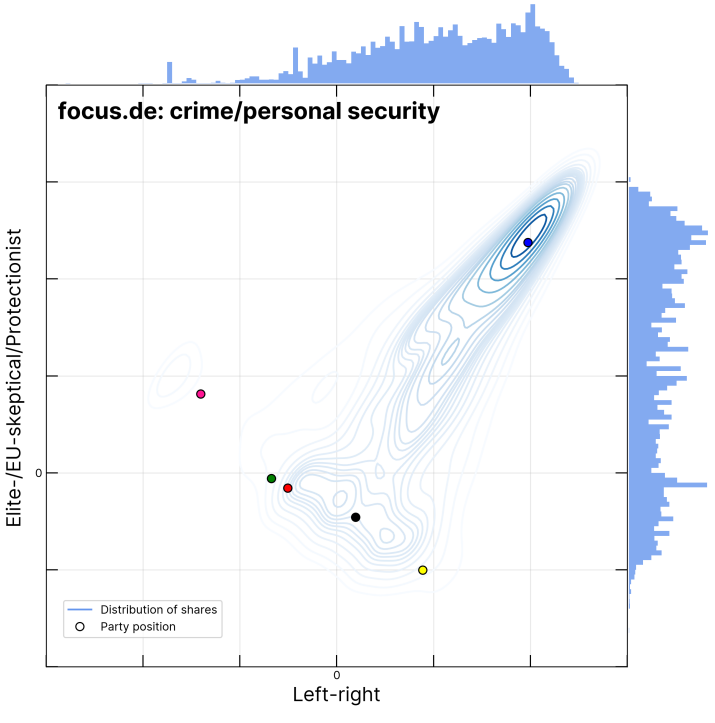

Figure 6: Distribution of news by focus.de on Crime/Personal Security.

Appendix K Other topics attributed to metatopic 'Environment and Climate Change'

See Fig. 7.

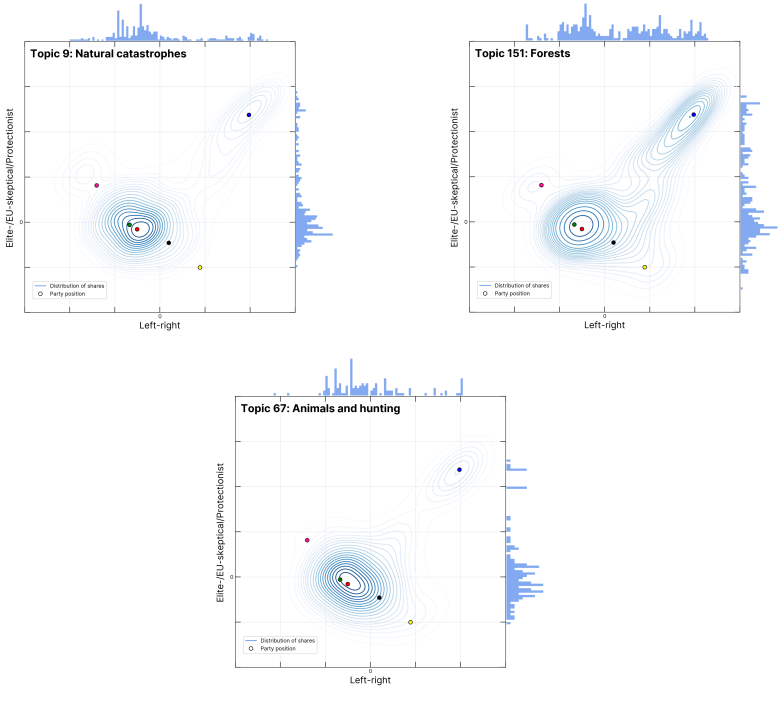

Figure 7: Other topics attributed to metatopic 'Environment and Climate Change'

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